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Chapter 6

Nonparametric tests: Kolmogorov–Smirnov and Peacock

6.1 Introduction

Nonparametric inference is applied when the user is not willing to limit the model of the distribution of the data to a specific parametric family. The modeling assumption of independence between observations is kept but the marginal distribution is not specified other than making the general requirement that this marginal distribution is continuous. According to this principle, in the context of testing the null hypothesis that the observations emerge from a specific distribution, the alternative hypothesis is composed of all other continuous distributions. In the two-sample setting if the null hypothesis is that the marginal distribution in both samples is the same then this common distribution may be any distribution.

Perhaps the most popular nonparametric test for univariate observations is the Kolmogorov–Smirnov test. In the one-sample setting this test compares the empirical cumulative distribution function of the data to the theoretical cumulative distribution function. The distance between the two functions is measured in terms of the largest discrepancy and the null hypothesis is rejected if this discrepancy is above a threshold of significance. In the two-sample case the two empirical cumulative distribution functions are compared with each other in terms of the largest discrepancy. Again, the null hypothesis of equality of the two distributions is rejected if this maximal discrepancy is larger than a threshold.

Peacock's nonparametric test is an extension to higher dimensions of the Kolmogorov–Smirnov test. Just like its one-dimensional counterpart, the Peacock test compares the empirical distribution function to an hypothesized distribution function in the single-sample version and in the two-sample version the empirical distribution in one sample is compared with the empirical distribution in the other sample. Distances between distribution functions are measured in terms of the largest discrepancy between the two functions. The null hypothesis that states that the underlying distribution of the observations is the hypothesized one (or is the same in both samples in the two-sample version) is rejected if that largest discrepancy exceeds a critical level.

In this section we will apply our method for the approximation of the tail distribution of extremes in a random field in order to identify the critical level for rejecting the null hypothesis. We will apply it first in the context of the univariate and one-sample Kolmogorov–Smirnov test. Later we will extend the proof to include the single-sample Peacock test in two-dimensions. Indications will be given regarding the potential extension of the proof to higher dimensions still and to the two-sample setting.

The analysis of the one-dimensional Kolmogorov–Smirnov test is classical. It appears in many textbooks and is presented as an example of nonparametric statistics in any decent above-introductory-level course on statistical theory. We will give momentarily a shortened version of that representation. Knowledge of the Peacock test is less wide spread. The analysis of this

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test is more complex since one can no longer use simplifications that are valid in the one-dimensional case but not in higher dimensions. Still, equipped with the interpretation of statistical statements as manifestation of random fields we may identify the similarity between the one-dimensional and the multi-dimensional goodness-of-fit tests. Viewing the problem as a random field will also help us analyze natural extensions of the nonparametric test. These extensions will be discussed in the last section.

6.1.1 Classical analysis of the Kolmogorov–Smirnov test

For the one-dimensional single-sample Kolmogorov–Smirnov test we consider a sequence of i.i.d. random variables $X_1, X_2, \ldots, X_n$. The empirical cumulative distribution function that is produced by these observations is $\hat{F}(t) = \frac{1}{n}\sum_{i=1}^{n} 1_{\{X_i \le t\}}$, for all real t. Under the assumption of continuous distribution we get that with probability 1 none of the observations share the same value. The empirical distribution function is flat between the ordered values of the observations and increases in jumps of size $1/n$ at these values. This empirical distribution is compared with the assumed underlying smooth distribution function $F(t)$. The discrepancy between the two functions is measured by the test statistic:

$$D = \sqrt{n}\max_t |\hat{F}(t) - F(t)| ,$$

and the null hypothesis is rejected for large values of D. The significance level of the test with a critical level y is given in terms of the probability $\mathrm{P}(D \ge y)$, where the probability is computed under the assumption that the actual marginal distribution of the random variables is indeed F.

The standard analysis proceeds by showing that the distribution of D is independent of F. This is justified by the application of the transformation F to each of the observations and to the sample space itself. This transformation produces a sample $U_i = F(X_i)$ of uniformly distributed random variables. The transformation of the sample space changes the theoretical cumulative distribution function $F(t)$ to the cumulative distribution function of the uniform distribution. The reflection of this transformation on the statistic D will move the locations of the jumps and change the cumulative distribution function $F(t)$ to the cumulative distribution function of the uniform distribution. The location where D obtains its maximal value will change but the maximal value of D will not. Consequently, the analysis can proceed assuming that the original observations were taken from the uniform distribution to begin with.

In the uniform case we have that $F(t) = t$, for $0 \le t \le 1$. The random process $\{\sqrt{n}[\hat{F}(t) - t] : 0 \le t \le 1\}$ has zero expectation and a variance-covariance function that is equal to

$$n\mathrm{Cov}(\hat{F}(t), \hat{F}(s)) = \mathrm{Cov}\left(1_{\{U_i \le t\}}, 1_{\{U_i \le s\}}\right) = \min(s,t) - s \cdot t ,$$

for all $0 \le s, t \le 1$. This corresponds to the expectation and covariance structure of the Brownian bridge. Indeed, from the theory of weak convergence of processes we get that the limit distribution, as n goes to infinity, of the given process is the Gaussian distribution that is identified by the expectation and covariance structure that we uncovered. The conclusion is that the limit distribution of D is the same as the distribution of the maximum of the absolute value of the Brownian bridge process.

The last part in the argument concludes with the analysis of the tail of the limit distribution of D. This tail is one of the few examples in which the distribution of the maximum of a random process has an explicit known form:

$$\mathrm{P}(D \ge y) \approx \mathrm{P}\left(\max_{0 \le t \le 1} |B_t| \ge y\right) = 2\sum_{k=1}^{\infty}(-1)^{k+1}e^{-2k^2y^2} \le 2e^{-2y^2} , \tag{6.1}$$

where $\{B_t : 0 \le t \le 1\}$ is the Brownian bridge process. The approximation in the above sequence of relations reflects the approximation of the re-scaled empirical process $\sqrt{n}[\hat{F}(t) - t]$ by its Brownian bridge limit. As a matter of fact, the validity

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of the inequality $\mathrm{P}(D \geq y) \leq 2e^{-2y^2}$, for any value of n and $y > 0$, was proved by Massart [13], strengthening a weaker version of the statement that goes under the title of the 'Dvoretzky–Kiefer–Wolfowitz inequality'. An alternative analysis, closer in spirit to our approach, is given in [14].

In the two-sample case the two empirical distribution functions are compared with each other using the statistic:

$$\sqrt{(n_1 n_2)/(n_1 + n_2)} \max_t |\hat{F}_1(t) - \hat{F}_2(t)| ,$$

for $\hat{F}_i$ the empirical distribution function produced from a sample of size n_i, $i = 1, 2$. The convergence to a Brownian bridge still holds, providing an asymptotic justification for using the function $2\exp\{-2y^2\}$ in order produce an approximation of the significance level of the test. However, the stronger version of the Dvoretzky–Kiefer–Wolfowitz inequality that was proved by Massart fails to hold in general [15].

6.1.2 Peacock's test

One of the goals of this chapter is to extend the analysis to the situation where the observations are multi-dimensional vectors rather than one-dimensional random variables. The statistic that we will consider was proposed by Peacock [16]. The statistic is similar to the statistic used for the Kolmogorov–Smirnov test: an empirical distribution function of the multivariate distribution of the vector is compared with a theoretical distribution in the one-sample case (or to another empirical distribution function in the two-sample case). A major difference between the two situations is that unlike the one-dimensional case, in the multi-dimensional case there is more than one natural candidate for an empirical distribution function that can be used in the comparison.

To clarify this point consider first a one-dimensional observation X_i. The evaluation of the empirical distribution that is associated with a point t on the real line corresponds to counting the number of times that the event $\{X_i \leq t\}$ occurred. Alternatively, one could have produced an empirical distribution by the consideration of events of the form $\{X_i \geq t\}$, since there is nothing special about the $-\infty$ end of the real line and distributions can be characterized by the probability above a point in the same way they are characterized by the probability below a point. However, in the one-dimensional setting we get that for either case the value of the statistic that measures the discrepancy between the resulting empirical distribution function and the theoretical probability is the same. Consequently, the Kolmogorov–Smirnov statistic does not depend on the orientation of the line.

The situation is different in the multivariate case. Specifically, consider the case where the observation $X_i = (X_{i1}, X_{i2})$ is a two-dimensional vector. A typical point in the plain is $t = (t_1, t_2)$. An empirical distribution that is associated with the point t can be constructed using events of the form $\{X_{i1} \leq t_1, X_{i2} \leq t_2\}$. However, different orientations for the two components within the vector may lead to the consideration of any one of the other three alternatives for constructing empirical distribution functions: $\{X_{i1} \geq t_1, X_{i2} \leq t_2\}$, $\{X_{i1} \leq t_1, X_{i2} \geq t_2\}$, or $\{X_{i1} \geq t_1, X_{i2} \geq t_2\}$. In this two-dimensional scenario the measure of discrepancy between the empirical distribution and its theoretical counterpart may differ, depending on the orientation that is used. Peacock's solution tothis problem is to use as a test statistic the largest of the four measures of discrepancy.

The analysis of the Peacock test is more complex than the analysis required for the Kolmogorov–Smirnov test for yet another reason. For the latter test there exists a transformation of the observations and the sample space that reduces the analysis to the analysis related to the uniform distribution and, at the same time, preserves the value of the statistic. This uniform distribution has some nice features that can be exploited in the derivation of the significance level. In the multivariate case, on the other hand,

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no such transformation exists. Consequently, the analysis should be conducted for the original distribution and the results of the analysis depend on this distribution.

Standard tools can still be used in order to produce an approximation for the significance level of the Peacock test. Indeed, it is not too difficult to show that the empirical field that is produced by considering the discrepancy between the empirical distribution function for a given orientation and the theoretical distribution function converges, after re-scaling, to a Gaussian random field. This field is parameterized by the points of the two-dimensional Euclidean space. The heuristic analysis of the tail properties of such field is given in Example J16 of [7]. The asymptotic expansion for the Peacock test may be obtained by summing the four orientation-specific approximations, each multiplied by 2 in order to account for both tails.

However, one may find this type of analysis less than satisfactory. First, it does not incorporate explicitly the effect of the sample size since the convergence to the Gaussian limit is conducted prior to the analysis of the tail distribution. And secondly, the analysis in Aldous' book [8] is heuristic. I have no doubt that the argument is provable. Yet, it is not clear to me how to add the rigor. The method that is based on the likelihood ratio identity, we hope, does not suffer from these drawbacks.

In the next section we reanalyze the Kolmogorov–Smirnov test using our method and produce a relation between the critical level of the test and its significance value. This application is an alternative to the classical analysis that was described above. It produces somewhat weaker results—asymptotic approximations rather than exact upper bounds—however, it has the advantage of being easier to generalize to higher dimensions as required for Peacock's test. In section 6.3 we consider the Peacock's multivariate extension of the Kolmogorov–Smirnov test. The discussion there will be brief since the argument is essentially the same argument that is used for the one-dimensional case. In section 6.4 we connect the discussion to the more general problem of multivariate scanning statistics.

6.2 Analysis of the one-dimensional case

Our goal in this section is to develop a proof of (6.1) as an approximation with the aid of the method that is based on the likelihood ratio identity.

To refresh our memory we may recall that the method initiates the analysis by formulating the problem at hand in terms of a random field. The leading term in the approximation is typically associated with the largest large-deviation rate of components of the field. The collection of parameter values can be restricted by the elimination of cases where the large-deviation rate trails off. Moreover, a discrete sub-collection of the restricted set of parameters values can be selected such that the distribution of the maximum of the original random field can be approximated by the maximum of the restricted and discrete sub-collection of elements. The analysis proceeds by the identification of an alternative measure and the use of the likelihood ratio identity to transform the problem of computing a tail probability to a problem that involves the sum of expectations, each of which is associated with a particular parameter value. These expectations can be approximated by the localization theorem. Finally, the approximations are integrated in order to produce the final formula.

6.2.1 Preliminary localization

A natural candidate for the random field in the one-dimensional setting is:

$$Y_t = \sqrt{n}[\hat{F}(t) - F(t)] = n^{-\frac{1}{2}} \sum_{i=1}^{n} [1_{\{X_i \le t\}} - \mathrm{P}(X_i \le t)] \,,$$

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defined for $-\infty < t < \infty$. The expectation of a component Y_t is equal to zero and the variance is equal to $F(t)[1 - F(t)]$. A normal approximation of the binomial distribution may trigger one to suspect that

$$\log \mathrm{P}(Y_t \geq y) \approx \log \bar{\Phi}(y/\{F(t)[1 - F(t)]\}^{\frac{1}{2}}) \approx -\frac{y^2}{2F(t)[1 - F(t)]} .$$

As a function of t, this first-order approximation of the marginal tail distribution of Y_t is maximized at the median $\hat{t} \Longleftrightarrow F(\hat{t}) = 0.5$, where the approximation of the marginal obtains the value e^{-2y^2}.

A refined analysis can be obtained by the examination of marginal probabilities in the vicinity of the median $\hat{t}$. The reduction in the exponent between the probability of the tail at t and the probability at $\hat{t}$ is:

$$\frac{y^2}{2}\left\{\frac{1}{1/4} - \frac{1}{F(t)[1 - F(t)]}\right\} = \frac{-2y^2(2F(t) - 1)^2}{1 - (2F(t) - 1)^2} \approx -8[f(\hat{t})]^2[y(t - \hat{t})]^2 .$$

Consequently, one may expect to be able to restrict maximization of the test statistic to the range $T = \hat{t} \pm C/y$, for some large but fixed value C, without a great loss in the resulting one-sided probability of exceeding the critical level.

A more careful analysis will inspect the large deviation rate for the centered and re-scaled binomial random variable Y_t as a function of both n and t instead of relying on the limit normal approximation. Here we will satisfy ourselves with the current heuristic argument until we introduce the appropriate log-moment generating function which can be used in order to identify the rates. In the final integration step we have a representation that includes a large deviation factor that is produced by these log-moment generating functions. These factors are the terms that should have been used in the argument that justified the preliminary localization step.

6.2.2 An approximation by a discrete grid

The second ingredient in the preliminary preparation for the application of the likelihood ratio identity is the approximation, based on a dense but finite sub-collection of values of the parameter, of the maximal value of the field. An attempt to mimic the analysis that was used in Section 3.2.2 for the scanning statistic will call for the analysis of an upper bound of the form:

$$e^{2y^2} \sum_{t \in \hat{T}} \mathrm{P}(Y_t \leq y - \frac{\epsilon}{y}, \max_{t \leq s \leq t + \delta/y^2} Y_s \geq y) , \tag{6.2}$$

where $\hat{T} \subset T$ is the dense sub-collection that is used in the approximation of the maximization over T.

An assessment of the bound in (6.2) can be carried out by the conditioning on the value of Y_t and an application of an inequality in the spirit of Fernique's inequality that is stated in Theorem 3.1. The proof of Fernique's inequality relies on a chaining argument and bounds on the tailsof marginal distributions of increments. Originally, the argument was applied to a Gaussian tail but the basic idea may be used for other types of tail behavior. A generalized argument is given in theorem [17] with their main result quoted in Theorem A.5 in the appendix. Unfortunately, we cannot use the theorem as stated since, as Chernoff's bound (Theorem A.4) demonstrates, the tail of the binomial distribution is sub-Gaussian with one order for the initial part of the tail and with a different order for the final part of the tail. Still, one can apply the chaining argument and obtain a useful bound on the tail of extreme values.

The minimal value of the random variable Y_t is $u_0 = -\sqrt{n}F(t)$. Denote $T_t = [t, t + \delta/y^2]$. Conditioning on the event $\{Y_t = u\}$, for u between u_0 and $y - \frac{\epsilon}{y}$, gives:

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$$\mathrm{P}\left(Y_t \le y - \frac{\epsilon}{y}, \max_{s\in T_t} Y_s \ge y\right) = \sum_{u=u_0}^{y-\frac{\epsilon}{y}} \mathrm{P}(Y_t = u)\mathrm{P}(\max_{s\in T_t} Y_s - Y_t \ge y - u | Y_t = u) \ .$$

This summation can be approximated by a Gaussian integration after the production of a bound on the conditional tail probability, since the probability mass function of Y_t can be approximated by a normal density. Some care should be given to small values of u, but for such values the conditional tail probability will be extremely small.

The maximum of the incremental process over the interval T_t can be bounded using a chaining argument. A basic ingredient for the application of this argument is a bound on the tail of increments:

$$Y_s - Y_r = \{N_{s,r} - nF_{s,r})\}/\sqrt{n} \ ,$$

for $N_{s,r}$ the count and $F_{s,r}$ the probability associated with the interval $(s, r]$, $s, r \in T_t$. It should be noted that the conditional distribution of $N_{s,r}$, given that $Y_t = u$, is binomial that counts the number of successes among $n_u = n(1 - F(t)) - u\sqrt{n} \approx n/2$ independent trials with a probability

$$F_{s,r|t} = [F(r) - F(s)]/[1 - F(t)] \approx 2f(t)(r - s)$$

of success. An increment can be represented in the form:

$$Y_s - Y_r = \{N_{s,r} - n_u F_{s,r|t})\}/\sqrt{n} - uF_{s,r|t} \ .$$

Observe that the conditional variance of an incremental count is:

$$\mathrm{Var}(N_{s,r}|Y_t = u) = n_u F_{s,r|t}(1 - F_{s,r|t}) \approx nf(t)(r - s) \ .$$

Consequently, if we apply the inequality (Theorem A.4) to an increment we get:

$$\mathrm{P}(Y_r - Y_s \ge x|Y_t = u) = \mathrm{P}(N_{s,r} - n_u F_{s,r|t} \ge \sqrt{n}(x + uF_{s,r|t})|Y_t = u)$$
$$\lesssim \exp\left\{-\frac{(x + 2uf(t)(r - s))^2}{2f(t)(r - s)} \frac{1}{1 + \frac{2u}{3\sqrt{n}} + \frac{x}{3\sqrt{n}f(t)(r-s)}}\right\} .$$

At the heart of the chaining argument is the representation of a typical increment $Y_s - Y_t$, for $s \in T_t$, as a telescopic sum:

$$Y_s - Y_t = \sum_{l=1}^{\infty}(Y_{t_l(s)} - Y_{t_{l-1}(s)}) \ ,$$

with $t_0(s) = t$ and $t_l(s)$ the central point of the interval of length $2^{-l}\delta/y^2$ in a partition of T_t that has 2^l elements. The interval that has $t_l(s)$ as a center covers the point s. Notice that $t_l(s) - t_{l-1}(s) = 2^{-(l+1)}\delta/y^2$ and denote $t_{j,l} = t + j\delta 2^{-l+1}/y^2$. Let $\pi_l = (1 - q)q^{l-1}$, for $1/\sqrt{2} < q < 1$. Observe that $y - u = \sum_{l=1}^{\infty}(y - u)\pi_l$.

An illustration of the chaining argument is given in Figure 6.1. It describes a pedigree that is constructed for the sake of producing a bound. The founder of the pedigree is located at point t. Each diagonal segment is associated with an increment between a parent and an offspring. The horizontal distance of the diagonal segment is the distance between the parameter value for the parent and the parameter value for the offspring. The vertical distance is the fraction of the total distance that is allocated for the increment between the parent and the offspring. In order for an increment between a point in the parameter space and the founder to exceed x at least one of the parent–offspring pairs must exceed the allocated distance. Hence, the probability that there is a point where the increment to the the founder is more than x is contained in the countably infinite union of events that a parent–offspring pair exceeds the allocated distance.

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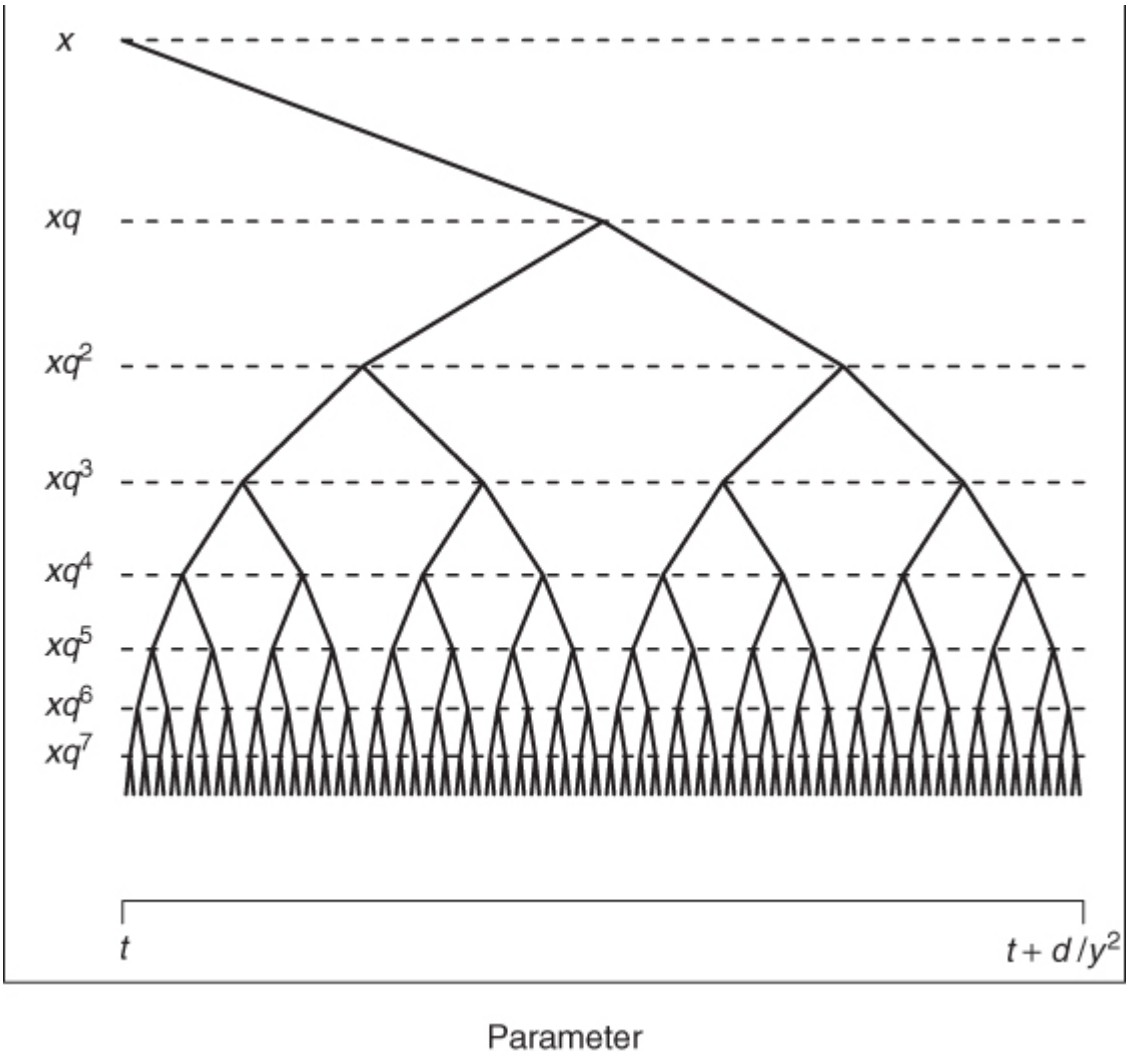
Figure 6.1 An illustration of the chaining argument. Diagonal segments correspond to a parent–offspring pair in a chain that connects any point with the founder of the pedigree.



It follows, from the application of the upper bound on the tail of an increment, that:

$$\begin{aligned}
&\mathrm{P}\left(\max_{s\in T_t} Y_s - Y_t \geq y-u \,|\, Y_t = u\right) \\
&\quad \leq \sum_{l=1}^{\infty}\sum_{j=1}^{2^{l+1}} \mathrm{P}\left(Y_{t_{j,l}} - Y_{t_{j-1,l}} \geq (y-u)\pi_l \,|\, Y_t = u\right) \\
&\quad \approx \sum_{l=1}^{\infty} 2^l \exp\left\{-\frac{(y(y-u)(q\sqrt{2})^l(1/q-1)+(u/y)f(t)2^{-\frac{l}{2}}\delta)^2}{\delta f(t)\left[1+\frac{2u}{3\sqrt{n}}+\frac{y^2(y-u)(2q)^l(1/q-1)}{1.5\sqrt{n}f(t)\delta}\right]}\right\}.
\end{aligned}$$

We complete the analysis by multiplying the bounds by $\exp\{2y^2\}$ times the density of Y_t and summation over $\hat{T}$. The summation is approximated by integration and a change of variable $x = y(y-u)$ is used in order to produce:

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$$(6.2) \lesssim \sum_{l=1}^{\infty} |\hat{T}| 2^l \int_{u_0}^{y-\frac{\epsilon}{y}} \exp\left\{4y(y-u) - \frac{(y(y-u)(q\sqrt{2})^l(1/q-1) + \frac{u}{y} f(t) 2^{-\frac{l}{2}} \delta)^2}{\delta f(t)\left[1 + \frac{2u}{3\sqrt{n}} + \frac{y^2(y-u)(2q)^l(1/q-1)}{1.5\sqrt{n} f(t)\delta}\right]}\right\} du$$

$$= \sum_{l=1}^{\infty} \frac{2C2^l}{\delta} \int_{\epsilon}^{y(y-u_0)} \exp\left\{4x - \frac{(x(q\sqrt{2})^l(1/q-1) + (1-x/y^2) f(t) 2^{-\frac{l}{2}} \delta)^2}{\delta f(t)\left[1 + \frac{2(1-x/y^2)}{3\sqrt{n}/y} + \frac{x(2q)^l(1/q-1)}{1.5\sqrt{n}/y f(t)\delta}\right]}\right\} dx \ ,$$

which converges to 0 when $\delta \to 0$.

6.2.3 Measure transformation

The alternative distributions that are used for the production of the likelihood ratio identity rely on the fact that Y_t is a linear transformation of binomial random variables. Recall that the log-moment generating function for the Bernoulli variable is $\varphi_p(\xi) = \log(pe^{\xi} + 1 - p)$, where p is the probability of success. It follows that the log-moment generating function for Y_t is

$$\psi_t(\xi) = n\left[\varphi_p\left(\xi/n^{\frac{1}{2}}\right) - \left(\xi/n^{\frac{1}{2}}\right)\varphi_p'(0)\right] \ ,$$

where $\varphi_p'(0) = p = p_t = F(t)$.

The large deviation rate is associated with the value of ξ that solves the equation

$$\psi_t'(\xi) = n^{\frac{1}{2}}\left[\varphi_p'\left(\xi/n^{\frac{1}{2}}\right) - \varphi_p'(0)\right] = y$$

$$\Rightarrow \quad \xi = \xi_t \approx y/\varphi_p''(0) = y/[F(t)(1-F(t))] \ .$$

The resulting approximation of the log of the marginal probability, the approximation that agrees to the first order with the normal approximation that we used above in order to restrict the parameter set to an interval about the median, takes the form:

$$\psi_t(\xi_t) - \xi_t\psi_t'(\xi_t) = n\left[\varphi_p\left(\xi_t/n^{\frac{1}{2}}\right) - \left(\xi_t/n^{\frac{1}{2}}\right)\varphi_p'\left(\xi_t/n^{\frac{1}{2}}\right)\right] \ .$$

The value of t that maximizes this quantity is denoted $\hat{t}$. Note that for $t = \hat{t}$ we obtain that $\xi_{\hat{t}} \approx 4y$.

Let T now be the grid with span δ/y^2 of points in the interval $\hat{t} \pm C/y$. Associate with each $t \in T$ a log-likelihood ratio of the form $\ell_t = \xi_t Y_t - \psi_t(\xi_t)$. With the aid of such log-likelihood ratios one may produce the likelihood ratio identity:

$$\mathrm{P}\left(\max_{t\in T} Y_t \geq y\right) = \sum_{t\in T} \mathrm{E}_t\left(\frac{1}{\sum_{s\in T} e^{\ell_s}}; \max_{s\in T} Y_s \geq y\right) \ ,$$

where P_t is the alternative distribution that is associated with the log-likelihood ratio ℓ_t.

A local random field is produced by the consideration of differences between the log-likelihood ratio at t and the log-likelihood ratios at other parameter values. The components of the local field are:

$$\ell_s - \ell_t = \xi_s Y_s - \xi_t Y_t - (\psi_s(\xi_s) - \psi_t(\xi_t)) \ .$$

This local field serves to produce the summation and maximization statistics:

$$S_t = \sum_{s\in T} e^{\ell_s - \ell_t} \ , \quad \text{and} \quad M_t = \max_{s\in T} e^{\ell_s - \ell_t} \ .$$

Combine that with the re-centered log-likelihood ratio:

$$\tilde{\ell}_t = \xi_t(Y_t - \psi_t'(\xi_t)) \ ,$$

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in order to get a representation of the summands in the log-likelihood identity that is convenient for the application of Theorem 5.2:

$$\mathrm{E}_t\left(\frac{1}{\sum_{s\in T}e^{\ell_t}};\max_{s\in T}Y_s\geq y\right)\approx e^{\psi_t(\xi_t)-\xi_t\psi_t'(\xi_t)}\mathrm{E}_t\left(\frac{M_t}{S_t}e^{-[\tilde{\ell}_t+m_t]};\tilde{\ell}_t+m_t\geq 0\right),$$

where $m_t=\log M_t$.

6.2.4 The asymptotic distribution of the local field and the global term

The null distribution P assigns the marginal density f to the independent observations $X_1,\ldots,X_n$. The distribution P_t corresponds to an alternative distribution over the same collection of observations. The observations under the alternative distribution are again i.i.d., but the marginal density is tilted and takes the form $f_t(x)=f(x)\exp\{\xi_t(1_{\{x\leq t\}}-F(t))/n^{\frac{1}{2}}-\psi_t(\xi_t)/n\}$.

In order to apply the localization theorem we need to identify the local limit distribution of $\tilde{\ell}_t$ and of the local field $\{\ell_s-\ell_t:s\in T\}$ and prove asymptotic independence between them. The analysis of the limit distributions should be carried out in the context of the alternative distribution P_t. This analysis is simplified substantially by the fact that both the centered log-likelihood ratio $\tilde{\ell}_t$ and the local field are produced as linear functions of the collection of independent random fields:

$$\{1_{\{X_i\leq s\}}:s\in T\},\quad i=1,2,\ldots,n.$$

Due to this fact we are in a position to apply Theorem 5.3 and get that the limit joint distribution of the local field and the global term is Gaussian. Computation of the expectation and covariance structure are sufficient for the specification of the limit local field and the proof of asymptotic independence.

For the analysis of the local field one may examine the representation:

$$Y_s=\sum_{i=1}^{n}\left[1_{\{X_i\leq s\}}-F(s)\right]n^{-\frac{1}{2}}.$$

Let $y=y_n$ diverge to infinity with the increase in the sample size. The expectation of the sum Y_s is the sum of the expectations. Consequently, the expectation of a component in the sum under the alternative distribution is:

$$\begin{aligned}
&n^{-\frac{1}{2}}\mathrm{E}_t\left[1_{\{X_i\leq s\}}-F(s)\right]\\
&\quad=n^{-\frac{1}{2}}\mathrm{E}\left[\left(1_{\{X_i\leq s\}}-F(s)\right)e^{\frac{\xi_s}{\sqrt{n}}\left(1_{\{x\leq t\}}-F(t)\right)-\psi_t(\xi_s)/n}\right]\\
&\quad\approx\frac{4y}{n}\mathrm{E}\left[\left(1_{\{X_i\leq s\}}-F(s)\right)\left(1_{\{X_i\leq t\}}-F(t)\right)\right]\\
&\quad=\frac{4y}{n}\mathrm{Cov}\left(1_{\{X_1\leq s\}},1_{\{X_1\leq t\}}\right),
\end{aligned}$$

where the approximation is valid if $n\to\infty$ and $y/n^{\frac{1}{2}}\to 0$. In the approximation we used the fact that $e^x\approx 1+x$, for small x, and the fact that $\xi_s\approx 4y$. From this approximation we may conclude that for large values of n:

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$$\mathrm{E}_t\left[\xi_s Y_s - \xi_t Y_t)\right] \approx 16y^2\left[\mathrm{Cov}\left(1_{\{X_1\le s\}}, 1_{\{X_1\le t\}}\right) - \mathrm{Var}\left(1_{\{X_1\le t\}}\right)\right] .$$

The deterministic part of the local field is $-\psi_s(\xi_s) + \psi_t(\xi_t)$. However, for large values of n and y :

$$\psi_s(\xi_s) = n\left[\varphi_p\left(\xi_s/n^{\frac{1}{2}}\right) - \left(\xi_s/n^{\frac{1}{2}}\right)\varphi_p'(0)\right] \approx 16y^2\frac{1}{2}\mathrm{Var}\left(1_{\{X_1\le s\}}\right) ,$$

with a similar approximation holding for $\psi_t(\xi_t)$. Therefore,

$$\psi_s(\xi_s) - \psi_t(\xi_t) \approx 8y^2\left[\mathrm{Var}\left(1_{\{X_1\le s\}}\right) - \mathrm{Var}\left(1_{\{X_1\le t\}}\right)\right] .$$

Combining the approximation for the expectation of the random, the approximation of the deterministic part, and expression for the variance and covariance of Bernoulli random variables we obtain:

$$\begin{aligned}\mathrm{E}_t(\ell_s - \ell_t) &\approx 16y^2[F(s\wedge t) - F(s)F(t) \\ &\quad - \frac{1}{2}\{F(s)(1-F(s)) + F(t)(1-F(t))\}] \\ &= -\frac{1}{2}16y^2[F(s\vee t) - F(s\wedge t) - (F(s) - F(t))^2] .\end{aligned}$$

Specifically, for $s = t + j\delta/y^2$, with j that is at most slowly changing as a function of y , we get:

6.3 $\mathrm{E}_t(\ell_s - \ell_t) \approx -8f'(t)\delta|j| \approx -8f'(\hat{t})\delta|j|.$

The variance of $\ell_s - \ell_t$ involves a sum of variances. Straightforward computations, and the observation that the variances of components of the random field under the alternative distribution are asymptotic to the variance under the null distribution, will lead to the approximation:

$$\mathrm{Var}_t(\ell_s - \ell_t) \approx 16y^2[F(s\vee t) - F(s\wedge t) - (F(s\vee t) - F(s\wedge t))^2].$$

As a result we get that:

6.4 $\mathrm{E}_t(\ell_s - \ell_t) \approx -\frac{1}{2}\mathrm{Var}_t(\ell_s - \ell_t).$

The analysis of the covariance between $\ell_s - \ell_t$ and $\ell_r - \ell_t$ is similar and yields in the case $t < r < s$:

$$\mathrm{Cov}_t(\ell_s - \ell_t, \ell_r - \ell_t) \approx 16y^2[F(r) - F(t) - (F(s) - F(t))(F(r) - F(t))].$$

Letting $y \to \infty$ will unfold the fact that the asymptotic distribution of $\{\ell_s - \ell_t\}$, for $s = t + j\delta/y^2$ and $|j|$ not too large, is a two-sided Gaussian random walk with a negative drift. The variance of an increment of that random walk is $16f'(\hat{t})\delta$.

Under the alternative distribution we get that the expectation of $\tilde{\ell}_t$ is 0 and the variance is $16y^2\psi_t''(\xi_t) = 16y^2\varphi_p''(\xi_t/n^{\frac{1}{2}})$, for $p = F(t)$. If we take $\kappa = y^2$ for the purpose of Theorem 5.2 and if $y/n^{\frac{1}{2}} \to 0$ then we get that $\mu = 0$ and, since $t \to \hat{t}$ and thus $\varphi_p''(\xi_t/n^{\frac{1}{2}}) \to \varphi_{0.5}''(0) = 1/4$, that $\sigma^2 = 16/4 = 4$.

The validation of the asymptotic independence that is needed for Theorem 5.3 follows from the fact the the covariance between $\ell_s - \ell_t$ and $\tilde{\ell}_t$ is of the order of a constant. However, the standard deviation of $\tilde{\ell}_t$ diverges to infinity proportionally to y . Consequently, the correlation between the global term and elements of the local field tends to zero.

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6.2.5 Application of the localization theorem and integration

The two-sided Gaussian random walk with a negative drift (that equals half the variance) has emerged in previous examples. In particular, we encountered it in the case of a scanning statistic and in the context of the sequential test for a normal mean. In both cases, the limit of the expectation of the ratio between the maximum and the sum, the constant that summarizes the effect of local fluctuations on the probability that the random field will obtain a large value, is associated with the overshoot function ν that is approximated in (2.4):

$$\mathbf{6.5}\quad \mathrm{E}[\mathcal{M}/\mathcal{S}] = 8f'(\hat{t})\delta \cdot \nu\left([16f'(\hat{t})\delta]^{\frac{1}{2}}\right).$$

The conclusion of Theorem 5.2 in the current setting is that

$$\mathrm{E}_t\left(\frac{M_t}{S_t}e^{-[\tilde{\ell}_t+m_t]};\ \tilde{\ell}_t + m_t \geq 0\right) \approx \delta\frac{8f'(\hat{t})}{y\sqrt{8\pi}} \cdot \nu\left([16f'(\hat{t})\delta]^{\frac{1}{2}}\right),$$

which leads to the approximation:

$$\mathrm{P}\left(\max_{t\in T} Y_t \geq y\right) \approx \delta\frac{8f'(\hat{t})}{y\sqrt{8\pi}} \cdot \nu\left([16f'(\hat{t})\delta]^{\frac{1}{2}}\right)\sum_{t\in T} e^{\psi_t(\xi_t)-\xi_t\psi_t'(\xi_t)}$$

In order to obtain a formula for the situation where normal approximation is valid for the large variation components we may identify the limit of each exponent in the sum and then expand it as a function of t in the vicinity of $\hat{t}$:

$$\psi_t(\xi_t) - \xi_t\psi_t'(\xi_t) \approx -\frac{\xi_t^2}{2}\psi_t''(0) = -\frac{y^2}{2\psi_t''(0)},$$

since $\xi_t \approx y/\psi_t''(0)$. We use the fact that $\psi_t''(0) = F(t)[1-F(t)]$, the fact that

$$\begin{aligned} F(t)[1-F(t)] &= (0.5 + [F(t)-0.5])(0.5 - [F(t)-0.5]) \\ &= 0.25 - [F(t)-0.5]^2, \end{aligned}$$

and the fact that $F(t) - 0.5 = F(t) - F(\hat{t}) \approx f'(\hat{t})(t-\hat{t})$ to obtain:

$$\psi_t(\xi_t) - \xi_t\psi_t'(\xi_t) \approx -2y^2 - 8[f(\hat{t})]^2[y(t-\hat{t})]^2.$$

Upon plugging this approximation in the expression for the probability of and extreme value and setting $t = \hat{t} + j\delta/y^2$ we get:

$$\mathrm{P}\left(\max_{t\in T} Y_t \geq y\right) \approx e^{-2y^2} \cdot \nu\left([16f'(\hat{t})\delta]^{\frac{1}{2}}\right)\frac{2f'(\hat{t})\delta/y}{\sqrt{2\pi}}\sum_{j:|j|\leq yC/\delta} e^{-\frac{1}{2}[2f(\hat{t})j\delta/y]^2}.$$

The summation is approximated, for large values of y, by an integral of a Gaussian kernel which gives $2\Phi(C/[2f(\hat{t})]) - 1$, for Φ the standard normal cumulative distribution function. Letting $C \to \infty$ and $\delta \to 0$, and taking into account the fact that the limit of the function ν at the origin is equal to 1, one gets the approximation:

$$\mathrm{P}\left(\max_t Y_t \geq y\right) \approx e^{-2y^2}.$$

For large values of y the probability that the maximum of the absolute value exceeds y is asymptotic to twice the one-sided probability. Consequently, the resulting approximation is in agreement with the upper bound (6.1).

In the two-sample Kolmogorov–Smirnov test one examines the empirical distribution functions $\hat{F}_j(t) = (1/n_j)\sum_{i=1}^{n_j} 1_{\{X_{ij}\leq t\}}$, for $j = 1, 2$, the two samples. The test statistic maximizes the absolute value of:

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$$Y_t = \sqrt{\frac{n_1 n_2}{n_1 + n_2}}(\hat{F}_1(t) - \hat{F}_2(t)).$$

The maximal marginal probability again takes place at $\hat{t}$ the median of the common distribution function F. It is convenient to assume that the sample sizes are proportional to one another and both converge to infinity. Alternative measures are produced by tilting the statistics Y_t. Examination of local behavior of increments can be carried out via the representation:

$$Y_r - Y_s = \sqrt{\frac{n_1 n_2}{n_1 + n_2}}\left(\hat{F}_1(r) - \hat{F}_1(s) - F_{s,r}\right) - \sqrt{\frac{n_1 n_2}{n_1 + n_2}}\left(\hat{F}_2(r) - \hat{F}_2(s) - F_{s,r}\right) ,$$

and the bounding or approximation of each of the two terms. The entire derivation is a nice exercise for the reader to carry out.

6.2.6 Checking the conditions of the localization theorem

For the current example we can use the simplified version of Theorem 5.1. Still, just like we did for the two examples that were used for demonstration in Chapter 3, we still prefer to prove the conditions of the more complex Theorem 5.2.

In the context of the localization theorem we want to define a local σ-algebra and identify the local approximations of the terms that measure the effect of local fluctuations. It is convenient to use for that purpose the σ-algebra that is generated by the finite collection of increments: $\{\ell_s - \ell_t : |s - t| \le \tau/y^2\}$. Then $\hat{M}_t$ and $\hat{S}_t$ are the maximization and the summation, respectively, of likelihoods that are restricted to this finite collection.

Condition I^* holds trivially with $C = 1$. For Condition V^* we have that

$$\mathrm{P}_t\left(M_t > e^{g(y^2)}\right) = \mathrm{P}_t\left(\max_{\{s:|s-\hat{t}|\le C/y\}} e^{\ell_s - \ell_t} > e^{g(y^2)}\right) \le (2Cy/\delta)e^{-g(y^2)} ,$$

so taking $g(y^2) = (1+\epsilon)\log y$ will do the job. Conditions III^* and IV^* follow from the application of the local limit Theorem 5.3 so we are left only with the need to check Condition II^*.

Using the same argument that was used in the proof of Condition II^* for the examples that were analyzed in Chapter 3, and in particular the example of an indicator function serving as a kernel, we can see that it is sufficient to investigate the probabilities:

$$\mathrm{P}_t(\{\ell_s - \ell_t \ge \log(\epsilon p_s)\} \cap \{\tilde{\ell}_t + \hat{m}_t \in x + (0, \delta]\}) ,$$

for $|s - t| > \tau/y^2$. As in the case of the scanning statistic, we split the analysis between the case where $\tau/y^2 < |s - t| < (\log y)^2/y^2$ and the case where $(\log y)^2/y^2 < |s - t|$ and use uniform weights for the former region and exponentially decreasing weights for the latter region. More tedious details go into the proof but no new ideas.

6.3 Peacock's test

Peacock's test generalizes the Kolmogorov–Smirnov test to higher dimensions. The setting where this test can be applied involves a sample $X_1, \ldots, X_n$ of independent random points on a d-dimensional plane. Henceforth we will assume that $d = 2$. However, the generalization to a finite dimension that is larger than 2 is straightforward.

Denote the distribution of X_i on the plane by F and define the empirical distribution over Borell subsets of T:

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$$\hat{F}(A) = \frac{1}{n}\sum_{i=1}^{n} 1_{\{X_i \in A\}}.$$

The Peacock nonparametric goodness-of-fit test is based on the difference between the empirical and theoretical distribution functions, applied to different quadrants. Specifically, let $t = (t_1, t_2)$ be a point in the support of F and consider the four closed quadrants with respect to this point by:

$$Q_1(t) = (\infty, t_1] \times (\infty, t_2]\,, \quad Q_2(t) = (\infty, t_1] \times (-\infty, t_2]\,,$$
$$Q_3(t) = (-\infty, t_1] \times (-\infty, t_2]\,, \quad Q_4(t) = (-\infty, t_1] \times (\infty, t_2].$$

The re-scaled difference between the empirical distribution and the probability assigned to the set $Q_j(t)$ is $Y_{j,t} = \sqrt{n}\{\hat{F}(Q_j(t)) - F(Q_j(t))\}$. The test statistic is based on the largest discrepancy between these empirical distribution functions:

$$\max_{j\in\{1,2,3,4\}} \max_{t\in T} |Y_{j,t}|.$$

The null hypothesis of validity of the assumed distribution F is rejected if this statistic is larger than a critical level. If the critical level is equal to y we get that the significance level of the test is:

$$\mathrm{P}\left(\max_{j\in\{1,2,3,4\}} \max_{t\in T} |Y_{j,t}| \geq y\right) \approx 2\sum_{j=1}^{4} \mathrm{P}\left(\max_{t\in T} Y_{j,t} \geq y\right)\,,$$

where the approximation is appropriate for large values of y. Our goal is to produce an approximation for the probabilities that appear in the sum.

We will derive an approximation for the case $j = 1$. Hence, in the following we eliminate the index j and denote $\hat{F}(t) = \hat{F}(Q_1(t))$ and $F(t) = F(Q_1(t))$.

The random field in the current example is two-dimensional. The marginal distributions that are associated with elements Y_t of the field are binomial, the same as they were in the one-dimensional setting of the Kolmogorov–Smirnov test. Their large deviation rates can assessed via the examination of the terms

$$\psi_t(\xi_t) - \xi_t \psi_t'(\xi_t) = n\left[\varphi_p\left(\xi_t/n^{\frac{1}{2}}\right) - \left(\xi_t/n^{\frac{1}{2}}\right)\varphi_p'\left(\xi_t/n^{\frac{1}{2}}\right)\right]\,,$$

where $\varphi_p(\xi)$ is the log-moment generating function of the Bernoulli random variable with $p = F(t)$ the probability of success and $\xi_t \approx y/[p(1-p)]$ is the value ξ that solves the equation:

$$\psi_t'(\xi) = n^{\frac{1}{2}}\left[\varphi_p'\left(\xi/n^{\frac{1}{2}}\right) - \varphi_p'(0)\right] = y.$$

Again, we would like to consider maximization of this rate. This maximization occurs when $p = 1/2$ and leads to the same rate that is introduced in the one-dimensional case. However, unlike the one-dimensional case, the collection of t values that obtains this maximization is not a single point but, instead, it is a curve in the plane:

$$\hat{t} = \{t : F(t) = 1/2\}.$$

It should be noted that this curve is compact. Due to the implicit function theorem the curve is also smooth if the density of the underlying distribution is smooth.

The preliminary localization step narrows the range of interesting parameter values to a tube about this curve:

$$\{t : |F(t) - 1/2| \leq C/y\}\,,$$

for a large enough C.

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The approximation of the maximal statistic by the maximizer of a discrete two-dimensional field is obtained via the bounding of probabilities such as:

$$\mathrm{P}\left(Y_t \le y - \frac{\epsilon}{y}, \max_{s\in T_t} Y_s \ge y\right),$$

where currently $T_t = [t_1, t_1 + \delta/y^2] \times [t_2, t_2 + \delta/y^2]$. These probabilities are the translation to the two-dimensional setting of the probabilities that are given in the sum (6.2) that was analyzed in the one-dimensional case. Bounding this probability can be carried out by conditioning on the value of Y_t and producing a Fernique's type of inequality for the tail of the conditional incremental random field. The proof of the inequality itself follows the proof that was developed for the one-dimensional case word for word. The only difference here is that each parent point has four offspring instead of the two that are used in the one-dimensional scenario. The four offspring are located at the centers of the four quadrants that divide the square that has the parent as its center.

The fact that a partition to four quadrants is carried out changes the number of elements in the l th level of the partition from 2^l to 4^l. Another difference is the change in the definition of the probability $F_{r,s} = F(r) - F(s)$ from a probability of an interval to a probability of a slightly more complex set: $Q(r)\backslash Q(s)$. The approximation of this probability for r in the vicinity of s, is $F_{r,s} \approx \dot{F}(s)'(r-s)$, for $\dot{F}(s)$ the gradient of the cumulative distribution function, evaluated at the point s.

The alternative distribution and the measure transformation are unaffected by the change in dimensionality. The localization theorem can be quoted essentially verbatim from the proof that is developed for the one-dimensional case. A slight modification appears in the step that identifies the limit of the local field and the resulting evaluation of the term that measures the effect of local fluctuations. In the current setting one obtains for $s = (s_1, s_2) = (t_1 + j_1\delta/y^2, t_2 + j_2\delta/y^2)$ that the limit variance of the corresponding element of the local field is:

$$\mathrm{Var}(\ell_s - \ell_t) \approx 16\dot{F}_1(t)\delta|j_1| + 16\dot{F}_2(t)\delta|j_2|,$$

for $\dot{F}_1$ and $\dot{F}_2$ the first and second components of the gradient of F. The limit covariance structure of the local field suggests that limit of the local field corresponds to a sum of two independent two-sided random walks. These random walks have negative drifts that equal half the variances. The conclusion regarding the term that measures the contribution of local fluctuations is that:

$$\mathrm{E}[\mathcal{M}/\mathcal{S}] = \delta^2 \prod_{i=1}^{2} \left\{8\dot{F}_i(t)\cdot \nu\left([16\dot{F}_i(t)\delta]^{\frac{1}{2}}\right)\right\}. \tag{6.6}$$

Consequently, since the distribution of $\tilde{\ell}_t$ is unaffected by the change in dimensionality, the approximation that results from Theorem 5.2 is:

$$\mathrm{E}_t\left(\frac{M_t}{S_t}e^{-[\tilde{\ell}_t + m_t]}; \tilde{\ell}_t + m_t \ge 0\right) \approx \mathrm{E}[\mathcal{M}/\mathcal{S}]/(y\sqrt{8\pi}).$$

Finally, the conclusion from the integration step is that:

$$\mathrm{P}\left(\max_{t\in T} Y_t \ge y\right) \approx \prod_{i=1}^{2}\sum_{t\in T}\left\{8\dot{F}_i(t)\cdot \nu\left([16\dot{F}_i(t)\delta]^{\frac{1}{2}}\right)\right\}\cdot \frac{\delta^2}{y\sqrt{8\pi}}e^{\psi_t(\xi_t) - \xi_t\psi_t'(\xi_t)}.$$

With the increase of the threshold to infinity we get that the sum converges to an integral. An approximation of the original probability may be obtained by sending C to infinity and δ to zero.

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6.4 Relations to scanning statistics

The Peacock test is an example of analysis of a scanning statistic on a two-dimensional space. The type of signals that one attempts to discover are quadrants of the plane and the source of information and randomness is the empirical distribution. The same principles that were applied for the analysis of this test are relevant to other types of scanning statistics. The results of the analysis tend to share similarities.

An immediate extension of the Peacock test will consider empirical distributions of rectangular regions of the form $[t_1, s_1] \times [t_2, s_2]$. Such regions are parameterized by pairs of points (t, s) . Hence, the parameter space is a subset of $\mathbb{R}^4$. Maximizing the re-scaled absolute difference between the empirical distribution of rectangles and their theoretical probabilities, where the maximization is taken over the parameter space, forms the basis for an alternative goodness-of-fit test. This type of test is described as Example J17 in [7].

The analysis of this extension should be a simple exercise for the reader who completed the details for the different variants of the Kolmogorov–Smirnov and the Peacock tests that were discussed in the previous sections. Again, the exponent of the large deviation rate is associated with rectangles of probability 1/2 and is asymptotic to $-2y^2$, where y is the threshold. For the term that measures the local fluctuations we need to change the power in (6.6) from 2 to 4. Other minor details need to be modified, but the modifications aretrivial.

This last example has a parallel formulation in the context of scanning statistics that generalize the basic example of Section 2.3 to situations where the signal emerges in a two-dimensional, or more generally d -dimensional, space. One difference between this new generalization and the goodness-of-fit test is the source of randomness. In the scanning statistic setting the source can be modeled as a higher dimensional white noise, Gaussian or not, instead of an empirical distribution. The choice of the source will affect the final statement of the outcome, but will have less of an effect on the method for obtaining that statement. Again, the large deviation rate will be associated with values of the parameter that maximize the marginal variance assuming that the expectation was taken care of. If the signal is in the form of an indicator of a rectangle then the term that is associated with local fluctuations will take the form of an appropriate product of terms associated with one-dimensional scanning. If the signal is of the form of a smooth kernel then the field is smooth and the resulting term is associated with the determinant of the Hessian that is associated with derivatives of the covariance function.

Another difference will be shown if the scanning statistic is standardized before the maximization over the parameter space. This situation corresponds more directly with the scanning statistics of Section 2.3. In this case the variance is constant across the parameter space and equals 1. The resulting exponent of large deviation will be fixed in the Gaussian case (and equal to $-y^2/2$) and will be almost fixed in non-Gaussian cases that are asymptotically Gaussian. The constant that summarizes the local fluctuations will be different, as well as the polynomial terms. Explicit formulae are known for the smooth case and for rectangular kernels with different characteristics. The details for some examples can be found in [18]. For kernels with nonrectangular shapes that produce a nonsmooth local field, say circles, one may write formally the term that is associated with local fluctuation as the expected ratio of a maximal exponentiated field over the sum of the same exponentiated field. However, we do not know of an explicit formula for this term and leave it as a challenge for the ambitious reader.

An even more challenging problem is to analyze the scanning statistics that are associated with shapes that are less regular than rectangles or circles. With such shapes the dimension of the parameter space may grow, a situation which is challenging, but not hopeless, for the method that is presented in this book.

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